ORF 524 — Midterm Exam 1

Solutions

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Contents

1 Point Estimation (25 points) **1**

2 Hypothesis Testing (25 points) **4**

Notes:

- Duration of examination: 80 minutes.
  An additional 10 minutes are given to upload your solutions to Gradescope.
- The exam has two questions with five parts each.
  All parts are equally weighted (5 points), for a total of 50 points.
  Please provide as much detail as possible in your answers.
  Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
  A single PDF file must be submitted to Gradescope.
- One sheet of personal notes (both sides) is allowed during the examination.
  No other notes, books, electronic devices, AI systems, or outside assistance may be used.
  This is otherwise a closed-book and individual examination.

1 Point Estimation (25 points)

Let $X_1, \dots, X_n$ be i.i.d. with Lebesgue density

$$f(x;\theta) = \frac{2x}{\theta} \exp\Big(-\frac{x^2}{\theta}\Big)\mathbf{1}\{x > 0\}, \qquad \theta \in \Theta = (0,\infty).$$

You may use without proof that $\mathbb{E}_\theta[X_i] = \sqrt{\pi\theta}/2$ and $\mathbb{V}_\theta[X_i] = \theta(1-\pi/4)$. Let $\mathsf{Exponential}(\theta)$ denote the distribution with Lebesgue density $\theta^{-1}e^{-y/\theta}\mathbf{1}\{y>0\}$, which has mean θ and variance θ^2. If $Y_1, \dots, Y_n$ are i.i.d. $\mathsf{Exponential}(\theta)$, then $\sum_{i=1}^n Y_i \sim \mathsf{Gamma}(n,\theta)$, which has mean $n\theta$ and variance $n\theta^2$. Estimators are compared by their mean squared error, $\mathrm{MSE}_\theta(\widehat{\theta}) = \mathbb{E}_\theta[(\widehat{\theta}-\theta)^2]$. All limits are taken as $n \to \infty$.

1. Show that $T = \sum_{i=1}^n X_i^2$ is a sufficient statistic for θ, and that $X_i^2 \sim \mathsf{Exponential}(\theta)$.

Solution

The joint density of $(X_1, \dots, X_n)$ at $x = (x_1, \dots, x_n)$ factors as

$$\prod_{i=1}^n f(x_i;\theta) = \underbrace{\theta^{-n}\exp\big(-T(x)/\theta\big)}_{g(T(x),\theta)} \underbrace{\prod_{i=1}^n 2x_i\mathbf{1}\{x_i>0\}}_{h(x)}, \qquad T(x) = \sum_{i=1}^n x_i^2,$$

so T is sufficient by the factorization theorem. The support $(0,\infty)$ does not depend on θ, so the indicator belongs to h. For $y > 0$, since $X_i > 0$ almost surely,

$$\mathbb{P}_\theta(X_i^2 \le y) = \mathbb{P}_\theta(X_i \le \sqrt{y}) = \int_0^{\sqrt{y}} \frac{2x}{\theta} e^{-x^2/\theta}\,dx = \Big[-e^{-x^2/\theta}\Big]_0^{\sqrt{y}} = 1 - e^{-y/\theta},$$

and $\mathbb{P}_\theta(X_i^2 \le y) = 0$ for $y \le 0$. This is the $\mathsf{Exponential}(\theta)$ distribution function.

2. Find the maximum likelihood estimator $\widehat{\theta}_{\mathrm{ML}}$ of θ. Show that $\widehat{\theta}_{\mathrm{ML}}$ is unbiased, compute the Fisher information $\mathcal{I}_n(\theta)$ of the sample, and show that $\widehat{\theta}_{\mathrm{ML}}$ is a uniformly minimum variance unbiased (UMVU) estimator of θ.

Solution

The log likelihood and score are

$$\ell_n(\theta) = \sum_{i=1}^n \log(2X_i) - n\log\theta - \frac{T}{\theta}, \qquad \frac{\partial}{\partial\theta}\ell_n(\theta) = -\frac{n}{\theta} + \frac{T}{\theta^2} = \frac{n}{\theta^2}\Big(\frac{T}{n} - \theta\Big).$$

Since $T > 0$ almost surely, the score is positive for $\theta < T/n$ and negative for $\theta > T/n$, so ℓ_n has the unique global maximizer $\widehat{\theta}_{\mathrm{ML}} = T/n = n^{-1}\sum_{i=1}^n X_i^2$. By part 1, $\mathbb{E}_\theta[\widehat{\theta}_{\mathrm{ML}}] = \mathbb{E}_\theta[X_i^2] = \theta$, so $\widehat{\theta}_{\mathrm{ML}}$ is unbiased, and $\mathbb{V}_\theta[\widehat{\theta}_{\mathrm{ML}}] = \mathbb{V}_\theta[X_i^2]/n = \theta^2/n$.

The support does not depend on θ and $\{\mathbb{P}_\theta : \theta \in \Theta\}$ is a one-parameter exponential family, so the score and information identities hold. Using $\partial^2 \ell_n(\theta)/\partial\theta^2 = n/\theta^2 - 2T/\theta^3$,

$$\mathcal{I}_n(\theta) = -\mathbb{E}_\theta\Big[\frac{\partial^2}{\partial\theta^2}\ell_n(\theta)\Big] = -\frac{n}{\theta^2} + \frac{2n\theta}{\theta^3} = \frac{n}{\theta^2},$$

equivalently $\mathcal{I}_n(\theta) = \mathbb{V}_\theta[(T - n\theta)/\theta^2] = n\theta^2/\theta^4$. The Cramér–Rao lower bound for unbiased estimators of θ is therefore $1/\mathcal{I}_n(\theta) = \theta^2/n = \mathbb{V}_\theta[\widehat{\theta}_{\mathrm{ML}}]$ for every $\theta \in \Theta$. Since $\widehat{\theta}_{\mathrm{ML}}$ is unbiased and attains the bound at every θ, it is UMVU within the class covered by the bound. In an exponential family, differentiation under the integral sign is valid for every statistic with finite expectation, so this class contains every unbiased estimator with finite variance.

Alternative route (full credit). T is complete and sufficient because the model is a full rank one-parameter exponential family whose natural parameter $-1/\theta$ ranges over the open interval $(-\infty, 0)$. Since $\widehat{\theta}_{\mathrm{ML}} = T/n$ is an unbiased function of T, the Lehmann–Scheffé theorem gives the same conclusion.

3. Consider estimators of the form $\widehat{\theta}_{\mathrm{LIN}}(c) = cT$ with $c > 0$. Compute $\mathrm{MSE}_\theta(\widehat{\theta}_{\mathrm{LIN}}(c))$, and find the value $c^\star$ that minimizes it. Show that $\mathrm{MSE}_\theta(\widehat{\theta}_{\mathrm{LIN}}(c^\star)) < \mathrm{MSE}_\theta(\widehat{\theta}_{\mathrm{ML}})$ for every $\theta \in \Theta$. Is $\widehat{\theta}_{\mathrm{ML}}$ admissible under squared error loss? Explain why your answer does not contradict part 2.

Solution

Since $T \sim \mathsf{Gamma}(n, \theta)$, $\mathbb{E}_\theta[cT] = cn\theta$ and $\mathbb{V}_\theta[cT] = c^2 n\theta^2$, so the bias–variance decomposition gives

$$\mathrm{MSE}_\theta(\widehat{\theta}_{\mathrm{LIN}}(c)) = c^2 n\theta^2 + (cn-1)^2\theta^2 = \theta^2\big[c^2 n(n+1) - 2cn + 1\big].$$

The bracket does not depend on θ and is a convex quadratic in c, minimized at $c^\star = 1/(n+1)$ for every θ. Since $\widehat{\theta}_{\mathrm{ML}} = \widehat{\theta}_{\mathrm{LIN}}(1/n)$,

$$\mathrm{MSE}_\theta(\widehat{\theta}_{\mathrm{LIN}}(c^\star)) = \theta^2\Big[\frac{n}{n+1} - \frac{2n}{n+1} + 1\Big] = \frac{\theta^2}{n+1} < \frac{\theta^2}{n} = \mathrm{MSE}_\theta(\widehat{\theta}_{\mathrm{ML}}) \quad \text{for every } \theta \in \Theta.$$

Thus $\widehat{\theta}_{\mathrm{LIN}}(c^\star) = T/(n+1)$ dominates $\widehat{\theta}_{\mathrm{ML}}$, and $\widehat{\theta}_{\mathrm{ML}}$ is *inadmissible* under squared error loss.

There is no contradiction with part 2. UMVU optimality compares $\widehat{\theta}_{\mathrm{ML}}$ only with unbiased estimators, whereas admissibility compares it with all estimators. The estimator $\widehat{\theta}_{\mathrm{LIN}}(c^\star)$ is biased, with bias $-\theta/(n+1)$, but shrinking toward zero lowers the variance to $n\theta^2/(n+1)^2$, which more than offsets the squared bias $\theta^2/(n+1)^2$.

4. Show that $\sqrt{n}(\widehat{\theta}_{\mathrm{ML}} - \theta) \rightsquigarrow \mathcal{N}(0, \theta^2)$, and that $\sqrt{n}(\widehat{\theta}_{\mathrm{LIN}}(c^\star) - \theta)$ has the same limiting distribution. What does this imply about the improvement found in part 3?

Solution

The variables X_i^2 are i.i.d. with mean θ and finite variance θ^2, and $\widehat{\theta}_{\mathrm{ML}}$ is their sample average, so the central limit theorem gives $\sqrt{n}(\widehat{\theta}_{\mathrm{ML}} - \theta) \rightsquigarrow \mathcal{N}(0, \theta^2)$. Next,

$$\sqrt{n}(\widehat{\theta}_{\mathrm{LIN}}(c^\star) - \theta) = \sqrt{n}\Big(\frac{n}{n+1}\widehat{\theta}_{\mathrm{ML}} - \theta\Big) = \frac{n}{n+1}\sqrt{n}(\widehat{\theta}_{\mathrm{ML}} - \theta) - \frac{\sqrt{n}}{n+1}\theta.$$

Since the deterministic sequences satisfy $n/(n+1) \to 1$ and $\sqrt{n}\theta/(n+1) \to 0$, Slutsky's theorem gives $\sqrt{n}(\widehat{\theta}_{\mathrm{LIN}}(c^\star) - \theta) \rightsquigarrow \mathcal{N}(0, \theta^2)$. Equivalently, $\widehat{\theta}_{\mathrm{LIN}}(c^\star) - \widehat{\theta}_{\mathrm{ML}} = -\widehat{\theta}_{\mathrm{ML}}/(n+1) = O_{\mathbb{P}}(n^{-1}) = o_{\mathbb{P}}(n^{-1/2})$.

The dominance in part 3 is a finite sample, higher order effect. The two estimators are first order asymptotically equivalent, their MSE ratio $n/(n+1)$ tends to one, and both have asymptotic variance $\theta^2 = 1/\mathcal{I}_1(\theta)$, so both are asymptotically efficient. Finite sample admissibility and first order asymptotic efficiency answer different questions.

5. The method of moments estimator based on $\mathbb{E}_\theta[X_i] = \sqrt{\pi\theta}/2$ is $\widehat{\theta}_{\mathrm{MM}} = 4\overline{X}_n^2/\pi$, where $\overline{X}_n = n^{-1}\sum_{i=1}^n X_i$. Show that $\widehat{\theta}_{\mathrm{MM}}$ is consistent for θ, and find the limiting distribution of $\sqrt{n}(\widehat{\theta}_{\mathrm{MM}} - \theta)$. Which of $\widehat{\theta}_{\mathrm{ML}}$ and $\widehat{\theta}_{\mathrm{MM}}$ would you prefer in large samples, and why?

Solution

Let $\mu = \mathbb{E}_\theta[X_i] = \sqrt{\pi\theta}/2$, $\sigma^2 = \mathbb{V}_\theta[X_i] = \theta(1 - \pi/4) < \infty$, and $g(u) = 4u^2/\pi$, so that $\widehat{\theta}_{\mathrm{MM}} = g(\overline{X}_n)$ and $g(\mu) = \theta$. By the weak law of large numbers, $\overline{X}_n \to_{\mathbb{P}} \mu$, and since g is continuous, the continuous mapping theorem gives $\widehat{\theta}_{\mathrm{MM}} \to_{\mathbb{P}} g(\mu) = \theta$. By the central limit theorem, $\sqrt{n}(\overline{X}_n - \mu) \rightsquigarrow \mathcal{N}(0, \sigma^2)$. Since g is differentiable with $g'(\mu) = 8\mu/\pi = 4\sqrt{\theta/\pi} \neq 0$, the delta method gives

$$\sqrt{n}(\widehat{\theta}_{\mathrm{MM}} - \theta) \rightsquigarrow \mathcal{N}\big(0, g'(\mu)^2\sigma^2\big) = \mathcal{N}\Big(0, \frac{16\theta}{\pi} \cdot \theta\Big(1 - \frac{\pi}{4}\Big)\Big) = \mathcal{N}\Big(0, \theta^2\Big(\frac{16}{\pi} - 4\Big)\Big).$$

Since $16/\pi - 4 \approx 1.093 > 1$, the asymptotic variance of $\widehat{\theta}_{\mathrm{MM}}$ exceeds the efficiency bound $\theta^2 = 1/\mathcal{I}_1(\theta)$ attained by $\widehat{\theta}_{\mathrm{ML}}$ in part 4. The asymptotic relative efficiency of $\widehat{\theta}_{\mathrm{MM}}$ with respect to $\widehat{\theta}_{\mathrm{ML}}$ is $\pi/(16 - 4\pi) \approx 0.915$, so $\widehat{\theta}_{\mathrm{ML}}$ (or $\widehat{\theta}_{\mathrm{LIN}}(c^\star)$) is preferred. Unlike $\widehat{\theta}_{\mathrm{ML}}$, the estimator $\widehat{\theta}_{\mathrm{MM}}$ is not a function of the sufficient statistic T and does not use all the information in the sample.

2 Hypothesis Testing (25 points)

Let $X_1, \ldots, X_n$ be i.i.d. with Lebesgue density

$$f(x;\theta) = \exp\big(-(x-\theta)\big)\mathbf{1}\{x \geq \theta\}, \qquad \theta \in \Theta = \mathbb{R}.$$

You may use without proof that $\mathbb{E}_\theta[X_i] = \theta + 1$ and $\mathbb{V}_\theta[X_i] = 1$. Let $X_{(1)} = \min_{1\leq i\leq n} X_i$ and $\overline{X}_n = n^{-1}\sum_{i=1}^n X_i$, and let $\mathsf{Exponential}(1)$ denote the distribution with Lebesgue density $e^{-y}\mathbf{1}\{y > 0\}$. For fixed $\theta_0 \in \mathbb{R}$ and $\alpha \in (0,1)$, consider the testing problem

$$H_0 : \theta \leq \theta_0 \qquad \text{versus} \qquad H_1 : \theta > \theta_0.$$

Let Φ denote the standard Normal distribution function and $z_{1-\alpha} = \Phi^{-1}(1-\alpha)$. All limits are taken as $n \to \infty$.

1. Show that $X_{(1)}$ is a sufficient statistic for θ, and that $n(X_{(1)} - \theta) \sim \mathsf{Exponential}(1)$ for every $\theta \in \Theta$ and $n \geq 1$.

Solution

Since $\prod_{i=1}^n \mathbf{1}\{x_i \geq \theta\} = \mathbf{1}\{x_{(1)} \geq \theta\}$, the joint density factors as

$$\prod_{i=1}^n f(x_i;\theta) = \underbrace{e^{n\theta}\mathbf{1}\{x_{(1)} \geq \theta\}}_{g(x_{(1)},\theta)} \underbrace{\exp\Big(-\sum_{i=1}^n x_i\Big)}_{h(x)},$$

so $X_{(1)}$ is sufficient by the factorization theorem. The support indicator must be retained: it carries all of the joint dependence of the likelihood on θ and the data. For $t \geq 0$, independence gives

$$\mathbb{P}_\theta\big(n(X_{(1)} - \theta) > t\big) = \prod_{i=1}^n \mathbb{P}_\theta\Big(X_i > \theta + \frac{t}{n}\Big) = \prod_{i=1}^n e^{-t/n} = e^{-t},$$

which is the $\mathsf{Exponential}(1)$ survival function.

2. Consider the test that rejects H_0 if and only if $X_{(1)} > \theta_0 + c$, for a constant $c \geq 0$. Find c such that the test has size α.

Solution

Let $\beta_n(\theta) = \mathbb{P}_\theta(X_{(1)} > \theta_0 + c)$. For $\theta \leq \theta_0 + c$, part 1 gives

$$\beta_n(\theta) = \mathbb{P}_\theta\big(n(X_{(1)} - \theta) > n(\theta_0 + c - \theta)\big) = e^{-n(\theta_0 + c - \theta)},$$

which is increasing in θ. The size is the largest rejection probability over the composite

null, $\sup_{\theta \le \theta_0} \beta_n(\theta) = \beta_n(\theta_0) = e^{-nc}$. Setting $e^{-nc} = \alpha$ gives

$$c = c_n = \frac{\log(1/\alpha)}{n}.$$

The size is attained at the boundary θ_0; for $\theta < \theta_0$ the rejection probability is $\alpha e^{-n(\theta_0 - \theta)} < \alpha$.

3. Derive the power function $\beta_n(\theta)$ of the size α test in part 2 for all $\theta \in \Theta$, and show that the test is consistent, that is, $\beta_n(\theta) \to 1$ for every fixed $\theta > \theta_0$. By inverting this family of tests indexed by θ_0, construct a confidence set for θ with exact coverage $1 - \alpha$.

Solution

By part 2 with $c_n = \log(1/\alpha)/n$, and because $X_{(1)} \ge \theta$ almost surely, so that $\beta_n(\theta) = 1$ whenever $\theta \ge \theta_0 + c_n$,

$$\beta_n(\theta) = \min\left\{1,\ \alpha e^{n(\theta - \theta_0)}\right\} = \begin{cases} \alpha e^{n(\theta-\theta_0)} & \text{if } \theta < \theta_0 + \log(1/\alpha)/n, \\ 1 & \text{if } \theta \ge \theta_0 + \log(1/\alpha)/n. \end{cases}$$

The power increases exponentially from 0 as $\theta \to -\infty$, equals α at θ_0, and reaches 1 at $\theta_0 + \log(1/\alpha)/n$. For any fixed $\theta > \theta_0$, $\beta_n(\theta) = 1$ as soon as $n \ge \log(1/\alpha)/(\theta - \theta_0)$, so $\beta_n(\theta) \to 1$ and the test is consistent.

The test of $H_0 : \theta \le \theta_0$ does not reject if and only if $X_{(1)} \le \theta_0 + \log(1/\alpha)/n$. Collecting the values θ_0 that are not rejected gives

$$C_n = \left\{\theta_0 \in \mathbb{R} : X_{(1)} \le \theta_0 + \frac{\log(1/\alpha)}{n}\right\} = \left[X_{(1)} - \frac{\log(1/\alpha)}{n},\ \infty\right),$$

and by part 1, $\mathbb{P}_\theta(\theta \in C_n) = \mathbb{P}_\theta(n(X_{(1)} - \theta) \le \log(1/\alpha)) = 1 - \alpha$ for every θ and n. Because $X_{(1)} \ge \theta$ almost surely, intersecting C_n with $(-\infty, X_{(1)}]$ does not change the coverage and yields the exact interval $[X_{(1)} - \log(1/\alpha)/n,\ X_{(1)}]$, whose length is $\log(1/\alpha)/n$.

4. Let $W_n = \sqrt{n}(\overline{X}_n - 1 - \theta_0)$, and consider instead the test that rejects H_0 if and only if $W_n > z_{1-\alpha}$. Show that this test has limiting size α, that is,

$$\lim_{n \to \infty} \sup_{\theta \le \theta_0} \mathbb{P}_\theta(W_n > z_{1-\alpha}) = \alpha.$$

Solution

Let $Z_n(\theta) = \sqrt{n}(\overline{X}_n - 1 - \theta)$. Under $\mathbb{P}_\theta$, each $X_i - \theta$ has density $e^{-u}\mathbf{1}\{u \ge 0\}$ for every θ, so $Z_n(\theta)$ has a distribution function G_n that does not depend on θ. Since

$W_n = Z_n(\theta) + \sqrt{n}(\theta - \theta_0)$ and G_n is nondecreasing,

$$\mathbb{P}_\theta(W_n > z_{1-\alpha}) = 1 - G_n\big(z_{1-\alpha} + \sqrt{n}(\theta_0 - \theta)\big) \le 1 - G_n(z_{1-\alpha}) \qquad \text{for } \theta \le \theta_0,$$

with equality at $\theta = \theta_0$. Thus $\sup_{\theta \le \theta_0} \mathbb{P}_\theta(W_n > z_{1-\alpha}) = 1 - G_n(z_{1-\alpha})$. The X_i are i.i.d. with mean $\theta + 1$ and variance 1, so the central limit theorem gives $Z_n(\theta) \rightsquigarrow \mathcal{N}(0,1)$, and since Φ is continuous, $1 - G_n(z_{1-\alpha}) \to 1 - \Phi(z_{1-\alpha}) = \alpha$. No Studentization is needed because $\mathbb{V}_\theta[X_i] = 1$ is known. Full credit for the pointwise limit at θ_0 combined with the monotonicity of the rejection probability in θ.

5. Consider the local alternatives $\theta_n = \theta_0 + h/\sqrt{n}$ with $h > 0$. Find $\lim_{n\to\infty} \mathbb{P}_{\theta_n}(\text{reject } H_0)$ for the test in part 2 and for the test in part 4. Then repeat both calculations for $\theta_n = \theta_0 + h/n$. Interpret the results in terms of the convergence rates of $X_{(1)}$ and $\overline{X}_n$.

Solution

Test based on $X_{(1)}$. By part 3, $\beta_n(\theta_0 + h/\sqrt{n}) = \min\{1, \alpha e^{h\sqrt{n}}\}$, which equals 1 once $\sqrt{n} \ge \log(1/\alpha)/h$, so the limit is 1. For $\theta_n = \theta_0 + h/n$, $\beta_n(\theta_n) = \min\{1, \alpha e^h\}$ for every n, which lies strictly between α and 1 when $h < \log(1/\alpha)$ and equals 1 otherwise.

Test based on $\overline{X}_n$. Under $\mathbb{P}_{\theta_n}$, write $W_n = Z_n(\theta_n) + \sqrt{n}(\theta_n - \theta_0)$, where $Z_n(\theta_n)$ has distribution function G_n as in part 4 and hence $Z_n(\theta_n) \rightsquigarrow \mathcal{N}(0,1)$. For $\theta_n = \theta_0 + h/\sqrt{n}$, $W_n = Z_n(\theta_n) + h \rightsquigarrow \mathcal{N}(h,1)$, so

$$\mathbb{P}_{\theta_n}(W_n > z_{1-\alpha}) \to 1 - \Phi(z_{1-\alpha} - h) \in (\alpha, 1).$$

For $\theta_n = \theta_0 + h/n$, $W_n = Z_n(\theta_n) + h/\sqrt{n} \rightsquigarrow \mathcal{N}(0,1)$ by Slutsky's theorem, so the rejection probability tends to $1 - \Phi(z_{1-\alpha}) = \alpha$, no better than the size.

Interpretation. $X_{(1)} - \theta = O_{\mathbb{P}}(n^{-1})$ with a non-Gaussian (exponential) limit, whereas $\overline{X}_n - 1 - \theta = O_{\mathbb{P}}(n^{-1/2})$ with a Gaussian limit. A test has nontrivial local power against alternatives at the distance that matches the rate of its statistic. The $X_{(1)}$ test detects alternatives at distance h/n and has power tending to one at distance $h/\sqrt{n}$, while the $\overline{X}_n$ test detects only alternatives at distance $h/\sqrt{n}$. Because the support depends on θ, the regular $\sqrt{n}$ Gaussian theory does not describe the attainable precision: the sufficient statistic $X_{(1)}$ is n consistent, whereas the moment based $\overline{X}_n - 1$ is only $\sqrt{n}$ consistent and discards the information in the support boundary.